LESSON PLAN

AIRCRAFT ARMAMENT (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Basic principle of operation of I-band Transponder
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Basic principle of operation of I-band Transponder. Frequency used. Interrogation and response format.
5. Specific Objectives	18.1.4.1 Trainee will be able to appraise the function of secondary radars in ASW operation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of I-band transponder interrogation and response formats using L & Ds and IB; Safety: Ensure no unauthorized transmission; Supervision: Instructor monitors PPT and PDF usage.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a secondary radar system?
(b) Which frequency band is used for I-band transponders?
(c) How does a transponder differ from a primary radar reflector?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principle of I-band transponder operation • Frequency allocation for interrogation and response • Interrogation pulse structure and modes • Response pulse format and data encoding • Role in Anti-Submarine Warfare operations	L & Ds	PPT, IB	15 min
(b)	• Detailed analysis of interrogation pulse sequences • Response pulse timing and modulation techniques • Integration of transponder data in ASW tracking • Common failure modes and troubleshooting basics • Operational procedures for transponder activation	L & Ds	PPT, PDF	20 min
(c)	• Summary of interrogation and response formats • Key frequency values and channel assignments • Transponder function in secondary radar ASW • Review of operational limitations and constraints • Preparation for next lesson on transponder roles	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the transponder responds to an interrogation pulse in ASW scenarios.
(b) Identify the specific frequencies used for I-band transponder interrogation and response.
(c) Describe the format of the response pulse generated by the transponder.

12. Home Assignment and Follow Up	Prepare a brief note on the specific role of the I-band transponder in Anti-Submarine Warfare operations to be discussed in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Role of I-band transponder. In ASW Operation.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min